

Cauchy Distribution

A continuous r.v. X is said to follow a Cauchy distribution if the pdf of X is as follows:

$f(x) = \frac{\lambda}{\pi[\lambda^2 + (x - \mu)^2]}$, $-\infty < x < \infty$, $\lambda > 0$ and is denoted as $X \sim C(\lambda, \mu)$ where λ, μ are two parameters. If $\lambda = 1$ and $\mu = 0$ then the r.v. X is said to follow standard Cauchy distribution, and the pdf of x is $f(x) = \frac{1}{\pi[1 + x^2]}$, $-\infty < x < \infty$

In general the moments and moment generating function of Cauchy distribution does not exist. But if it is assumed that the mean of the Cauchy distribution exist then it will be equal to μ and its mean, median and mode is same, i.e., μ .

Additive Property of Cauchy distribution

If r.v.s X_1, X_2 are independent r.v.s, with parameters (λ_1, μ_1) and (λ_2, μ_2) respectively then $X_1 + X_2$ will follow Cauchy distribution with parameter $(\lambda_1 + \lambda_2, \mu_1 + \mu_2)$